

Figure 1: A typical virtualisation architecture

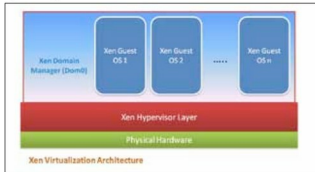


Figure 2: Xen architecture

There are three high-level types of virtualisation. The first is full virtualisation (also called a hypervisor), wherein all guest instances are kept separate from each other. In the second type, called para-virtualisation, the guest systems know about each other, which is achieved by the semi-abstraction of the physical and OS layers. The benefit of this method is that it takes less of a toll on the system's resources, resulting in faster guests, but compromises on security. The third type is called OS level virtualisation, in which the abstraction layer simulates multiple instances of the host OS. In this method, it is not possible to host a variety of operating systems. Hypervisor is the most commonly used method, and is not only limited to servers but also to networks, desktops, storage networks, applications and databases.

The main purpose behind virtualisation is to reduce hardware costs and increase ROI (returns on investment). It has also been observed that modern data centres are using virtualisation to reduce real estate costs, as well as to cut heat emissions from the infrastructure, in order to have eco-friendly (green) data centres. While there are multiple advantages of virtualisation, this technology is also plagued by cyber security problems in the physical infrastructures. Let us now explore a famous open source virtualisation solution and discuss various security challenges.

Xen open source virtualisation

I chose the Xen platform as an example solely because it is a famous and versatile solution, and is being used by major

cloud hosting players such as Amazon. It is a highly scalable platform, which enables multiplying the given physical hardware into powerful virtual instances, and is also backed by industry leading open source communities in terms of support and solutions engineering.

Technically speaking, Xen virtualisation has a hypervisor component at its heart, which is a software layer that runs between the physical hardware and the operating system to be able to create multiple guest instances. This layer also distributes and schedules the CPU time, partitions memory spaces, and performs seamless process multiplexing. As shown in Figure 2, the Xen architecture contains three components—the hypervisor layer, the domain manager and guest instances.

Xen domain manager is a kernel component which gets installed as the first domain, and it is a virtual instance by itself. It has special and elevated rights and access to the physical resources. The domain manager runs before any other guest instance can be started, and provides functions such as allowing the creation of child domains, creating and managing virtual devices for guest domains, and also to control network and disk resources to ensure fair resource allocation and utilisation by the guest instances. Guest instances, also called underprivileged DomainU components, are not allowed to access hardware resources directly. Guests communicate to domain managers through proxy calls to ask for access, and also keep the manager posted about their online or offline status. As for the benefits, besides cutting costs, the virtualisation solutions are used as a sand box to test the software code before it is deployed into the production environment. Xen is also effective for server migrations, wherein the virtual server is simply copied over to another powerful physical hardware platform, where it works without a glitch.

Securing Xen

As mentioned earlier, though Xen virtualisation comes with a lot of benefits, it can be susceptible to cyber security challenges if the hypervisor technology is not carefully studied. It is very important to keep in mind that virtualisation is a software component and can be affected by programmatic as well as man-made cyber attacks, just as any physical server infrastructure. In fact, virtualisation brings along more security challenges than physical hardware. The reasons for this are listed below.

Hypervisor security: Since this is a layer between the physical hardware and the operating system, it is bound to have vulnerabilities that can directly impact security. Like most operating system patches, the Xen platform provides timely service patch updates and security hot fixes to repair the bugs or vulnerabilities as they occur. The same goes for the domain manager too. During the installation phase, a tested and well written security patching process needs to be put in place, to ensure continuous security improvements.

(Continued on Page 86....)